Vaishnavi V

Software Developer

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Professional Summary

Aspiring AI/ML Engineer with strong foundations in Machine Learning, Deep Learning, Natural Language Processing, Computer Vision, and Cyber Security. Experienced in developing intelligent applications using Python, YOLOv5, OpenCV, TensorFlow, Streamlit, Flask and Java through academic and internship projects. Passionate about building secure, scalable, and real-world AI solutions.

Education

SRM Valliammai Engineering College — Graduated: 2026
B.Tech Information Technology — CGPA: **8.95/10**
St. John's Matriculation Higher Secondary School (HSC) — 89.6%
St. John's Matriculation Higher Secondary School (SSLC) — 75%

Skills

Programming: Python, Java
AI/ML: Machine Learning, Deep Learning, NLP, Computer Vision
Libraries: TensorFlow, Keras, OpenCV, YOLOv5, Scikit-learn, Transformers
Tools: Git, GitHub, Roboflow, VS Code, Google Colab

Experience

AI Intern – Retech Solutions — May 2025 – Jun 2025

- Developed AI and Computer Vision applications using Python, OpenCV and YOLOv5.
- Collected, annotated and preprocessed datasets using Roboflow for custom object detection models.
- Trained, evaluated and optimized deep learning models to improve detection accuracy.
- Automated testing workflows and strengthened skills in Machine Learning, Computer Vision and workflow automation.

Projects

SecureNote: Gesture-Based Smart Fake Currency Detector

- Developed an AI-powered fake currency detection system for visually impaired users.
- Integrated gesture-based interaction with a Flask web interface for improved accessibility.
- Trained a Vision Transformer (ViT) model using Roboflow datasets to classify genuine and counterfeit Indian currency.
- Enabled real-time voice feedback using pyttsx3 for accessibility.

EduSpeak: Bridging Education and Regional Languages

- Developed an AI-powered educational platform that extracts text from PDF documents.
- Implemented BART Transformer for automatic text summarization.
- Integrated translation APIs to convert educational content into Tamil.
- Generated speech output using gTTS and deployed the complete workflow through Streamlit.

Privacy-Preserving Web Login Framework Based on Zero-Knowledge Proofs

- Designed a passwordless authentication framework using Zero-Knowledge Proofs (ZKP).
- Implemented a secure challenge-response protocol that verifies user identity without revealing sensitive credentials.
- Eliminated password storage and transmission to defend against phishing, replay attacks, credential theft and database breaches.
- Enhanced both security and usability through privacy-preserving authentication.

Certifications

- Java Full Stack Development – CodeBuilders (2024)
- HTML, CSS & JavaScript – CodeBuilders (2024)
- Python Programming – K'Log Institution (2023)